careMESH

careMESH Inc
12110 Sunset Hills Road
Reston, VA 20194
(703) 691-6128

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The Honorable Thomas Keane, MD, MBA
Assistant Secretary for Technology Policy
Office of the National Coordinator for Health Information Technology
Department of Health and Human Services
Centers for Medicare & Medicaid Services
Mail Stop C4-26-05, 7500 Security Boulevard,
Baltimore, MD 21244–1850

RE: HEALTH TECHNOLOGY ECOSYSTEM

REFERENCE: CMS-0042-NC

Dear Assistant Secretary Thomas Keane,

Thank you for the opportunity to provide feedback on the request for information titled “Health Technology Ecosystem”.

We believe that the policy framework established over nearly three decades by the administration, CMS, and ASTP/ONC rulemakings has resulted in significant progress toward the goal of a seamless and secure flow of health information between patients, providers, and payers, and is beginning to enable digital workflows across the entire healthcare system.

We welcome the focus of this RFI -- to surface and help identify real-world challenges that further policy work would efficiently and effectively address. Pervasive, low-friction digital communication combined with increasingly powerful computing capabilities allows for the overarching goal to be more aggressive and optimistic than previous statements of the quadruple aim.

The goal is ***to continually improve health quality, efficiencies, and reduce costs for all, while enhancing the experience and satisfaction of both patients and providers.***

In this document, we provide recommendations in three areas: provider directory, direct secure messaging, and FHIR access to patient data.

I. CAREMESH CREDENTIALS TO OFFER FEEDBACK

careMESH offers this feedback based on our many years of experience developing and deploying: A) a National Provider Directory (NPD) based on the HL7 FHIR standard, B) a digital communications and message delivery service leveraging numerous ONC standards, and C) a care coordination platform that integrates with EHRs and leverages A and B.

1. A FHIR NPD: Six years ago, careMESH launched the nation's first National Provider Directory built entirely on the FHIR standard (see details here). Today, the directory is the definitive source of truth for our healthcare customers, containing verified data on over 8 million providers, organizations, and their locations.

Our platform creates a single, unified "golden record" for each provider by integrating hundreds of sources, including:

- Federal Datasets: CMS sources include NPPES, PECOS, and Medicare Compare.
- Proprietary & Commercial Sources: Hundreds of other non-federal & private data feeds.

Using sophisticated algorithms to normalize and integrate information, the directory provides comprehensive profiles for physicians, nurse practitioners, physician assistants, and millions of other licensed medical professionals. Among many other things, these profiles encompass organizational affiliations, practice locations, and contact information, including Direct Addresses and FHIR end-points to support our qualified customer's needs for reliably exchanging health information. Using native Master Data Management functionality, each customer can also use the platform to organize, enhance and present their own unique data in their own context(s), to support their own sophisticated and individual use cases.

Trusted by hospitals, health plans, health information exchanges, public health organizations, and others, the careMESH NPD is a versatile and secure solution. It is deployed as a cloud application serving millions of FHIR API requests, as a native EHR app, and as a standalone web app.

The platform and associated applications are both NIST and HITRUST certified, ensuring the highest standards of security and compliance.

1. B Communications Automation: Launched in tandem with our National Provider Directory, careMESH CONNECT is an advanced messaging and automation service designed to ensure the reliable delivery of medical records, messages, notifications, and other Protected Health Information (PHI) to any provider in the United States (see details here).

The platform provides a single, streamlined solution for clinical communication by leveraging a comprehensive suite of digital health standards and protocols, including HL7, FHIR APIs, C-CDA, and Direct Secure Messaging. This enables healthcare organizations to securely and efficiently transmit PHI from their existing EHRs and other systems, eliminating the need for manual, analog processes such as faxing.

Over the past six years, careMESH CONNECT has successfully delivered millions of messages to many tens of thousands of providers. Successful delivery is guaranteed to every practice and provider in the USA. The service ensures information reaches the right destination, whether it's directly into a destination EHR or to specific care team members, including physicians, PAs, nurse practitioners, care coordinators, and office staff.

1. Care Coordination, Workflow, and Task Management: careMESH NAVIGATE (see details here) is a modern care coordination and workflow management tool that operates directly within customer Electronic Health Records (EHRs). It empowers care teams to manage patient populations, automate tasks, and collaborate seamlessly with other providers and care teams, regardless of location.

It has been proven in complex use cases, including referral management, longitudinal care management of chronic disease patient cohorts, improving throughput in interventional cardiology, and navigation for cancer patients, among many others. Deployments are yielding significant results, including higher referral capture rates, faster time to intervention, fewer lost-to-follow-up cases, and operational efficiencies.

II. FEEDBACK and RECOMMENDATIONS

National Provider Directory (NPD, NDH)

To streamline our feedback in this section, we reference our response to the 2022 NDH RFI (reference: CMS–0058–NC), which we still consider relevant (CMS 2022 NDH RFI/careMESH).

The summary in short form is as follows:

- CMS should rationalize NPESS and PECOS provider data and focus on ensuring that the existing data content, including Direct Addresses and FHIR Endpoints, is accurate and up-to-date. CMS should not attempt to collect or provide additional data without first perfecting this dataset. Additionally, CMS should keep the data open source and therefore not attempt to capture any proprietary or private information.

- To achieve this, the data should be collected electronically from EHRs, which in most cases already record basic user information, such as name, specialties, identifiers, practice locations, and contact details, including Direct Addresses and FHIR endpoints. Furthermore, this should be enforced through a new CEHRT requirement, which will move the burden from CMS itself and providers to the technology companies that support them.

- CMS should present the data through FHIR APIs and provide a notification mechanism, such as webhooks, to inform subscribers to the directory when updates have been made.

Rationale and Discussion:

There is a temptation to include broad and varying data types in an NDH/NPD. However, individual users in every organization need to leverage core directory information, combined with local and specialized information, for their particular use cases. And there are millions of use cases. The hard part is standardizing and inter-relating provider names, credentials, locations, organizations, services, and digital and analog communication addresses. This is where CMS should focus. Suppose CMS/ONC succeeds in making this core information available through systems, standards, and methods that naturally maintain the accuracy and up-to-date nature of the information. In that case, all the other directory use-cases can and will be addressed by the organizations and individuals themselves, with occasional assistance from appropriate third parties.

Direct Messaging

- Today, Direct Messaging primarily functions for the exchange of C-CDAs, already providing strong encryption, security, privacy, identity management, and oversight to ensure the system is not abused by promotional messaging or spam.
- However, the implementation of Direct (rather than the standard) should be modernized so that it retains all of the current strengths (above), addresses the reliability challenges that impact message delivery, and functions similarly to email. It should become the default mechanism to send messages, to and from individual caregivers across the hallway as well as across the country. All EHR Users should be able to send messages to any other EHR user, including attachments such as C-CDAs, PDFs, and other files, and users should have confidence that they will be delivered to the recipient's mailbox and seen. This is not the case today.
- The standards, HISPs, and the Direct Protocol itself support all of this, as do many EHR vendors. But not all EHR vendors have implemented Direct in the same way. Providers don't trust it as the mechanism for sending anything but CCDA's, hence the continued reliance on Fax.
- As with directory data, CEHRT should introduce new requirements that test the ability of EHRs to send and receive messages containing varied content, and ensure that the message is received and routed to the intended recipient. Incentives to drive providers to properly enable and configure these capabilities may also be needed.
- The work to enable increased flexibility, utility, and range for unstructured messages should not impede the current emphasis on also supporting structured messages and automation within the Direct messaging framework, which should be continued.

FHIR APIs

- Current regulations require EHR vendors and their customers to make FHIR APIs available to patient-facing applications. The same should be true of provider-facing applications.
- Some EHR vendors ,such as Epic (with Open Epic) and Oracle/ Cerner (with Code), already offer good FHIR API access to provider-facing applications. But not all do. This frequently frustrates organizations, for example, making it challenging for two different organizations to collaborate on the care of a mutual patient.

- A method of programmatically notifying API clients that new data is available, such as a callback URL, webhook or other notification hook, should be standardized, required and enforced via CEHRT and provider incentives or similar.
- Broader, easier, and more pervasive use and access to FHIR resources by providers and their business partners would enable numerous new solutions to emerge, supporting patient navigation and care coordination. This, in turn, would lead to better, more efficient, and higher-quality care, while also reducing the burden on patients.
- The EHR Customer should stay in control of which providers or provider applications have access to patient data. Still, the range of data types available should continuously grow, and the ability to enable these tools should become easier.

III. CONCLUSION

From our comments above, it is clear that we see great merit in CMS, ASTP, and ONC investing more time and energy in improving the extensive standards, policies, and regulatory framework for health communications and interoperability that have been established over nearly three decades.

We believe the current regulatory, policy, and standards framework created by ONC and related organizations has been excellent and that with a few strategic enhancements, the enhanced framework, coupled with new computational methodologies, puts the US on the cusp of enabling transformational change to the discovery, delivery and efficiency of healthcare itself.

Many organizations are ready and willing to support CMS in this initiative, including ours, and we look forward to learning more about the strategy CMS adopts.

Sincerely

Peter Tippett, MD, PhD
CEO, CareMESH Inc
ptippett+1@caremesh.com